



Model #2 — Decision Tree Regression

```
[79]: # Step 1 - Import the model
```

```
[80]: from sklearn.tree import DecisionTreeRegressor
```

```
[81]: # Step 2 - Create Model #2
```

```
[82]: model_2 = DecisionTreeRegressor(random_state = 42)
```

Step 3 — Train it

We already have our ML-ready data:

X_train_processed X_test_processed y_train y_test

```
[86]: model_2.fit(X_train_processed, y_train)
```

```
[86]: ▾ DecisionTreeRegressor ⓘ ?  
    ▸ Parameters  
    ▸ Fitted attributes
```

```
[87]: y_pred_2 = model_2.predict(X_test_processed)
```



Next — Evaluate Model #2

```
[88]: from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score  
import numpy as np
```

```
mae_2 = mean_absolute_error(y_test, y_pred_2)  
mse_2 = mean_squared_error(y_test, y_pred_2)  
rmse_2 = np.sqrt(mse_2)  
r2_2 = r2_score(y_test, y_pred_2)
```

```
print("Model #2 - Decision Tree Regression")  
print("MAE :", mae_2)  
print("MSE :", mse_2)  
print("RMSE :", rmse_2)  
print("R² :", r2_2)
```

```
Model #2 - Decision Tree Regression  
MAE : 43803.50581395349  
MSE : 5120584623.200582  
RMSE : 71558.26034218958  
R² : 0.6092376724956141
```